



American International University-Bangladesh (AIUB)

Department of Computer Science
Faculty of Science & Technology (FST)
Summer 25-26
Section: B

Software Quality Assurance and Testing

Waypoint (AI-Based Student Career Guidance Platform)

Report submitted By

SN	Student Name	Student ID
1	Muhammad Talha Islam	22-49599-3
2	Wafa	22-49542-3

Checked By Industry Personnel

Name: Wasif Zaman

Designation: SQA Team Lead

Company: Riseup Labs

Sign:

Date: 10-09-2026

Software Test Plan

for

Waypoint (AI-Based Student Career Guidance Platform) Test Plan

Version 1.0 approved

Prepared by: Muhammad Talha Islam & Wafa

American International University-Bangladesh (AIUB)

23 August 2026

Table of Contents

Revision History	4
1. TEST PLAN IDENTIFIER.....	5
2. REFERENCES.....	5
3. INTRODUCTION.....	5
Background to the Problem.....	5
Solution to the Problem.....	5
Testing Focus	5
4. REQUIREMENT SPECIFICATION	6
4.1 System Features.....	6
4.2 System Quality Attributes	11
4.3 System Interface	12
4.4 Project Requirements.....	19
5. FEATURES NOT TO BE TESTED.....	21
6. TESTING APPROACH	21
6.1 Testing Levels	21
6.2 Test Tools	23
7. TEST CASES/TEST ITEMS	23
8. ITEM PASS/FAIL CRITERIA	31
9. TEST DELIVERABLES	31
10. STAFFING AND TRAINING NEEDS.....	32
11. RESPONSIBILITIES	32
12. TESTING SCHEDULE.....	33
13. PLANNING RISKS AND CONTINGENCIES	33
14. APPROVALS.....	34
15. Meetings	34

Revision History

Revision	Date	Updated by	Update Comments
0.1	2026.08.23	Wafa	First Draft
0.2	2026.08.25	Muhammad Talha Islam	Second Draft
0.3	2026.08.31	Wafa	Third Draft
0.4	2026.09.04	Muhammad Talha Islam	Forth Draft
0.5	2026.09.09	Muhammad Talha Islam	Final Draft

1. TEST PLAN IDENTIFIER

Waypoint (AI-Based Student Career Guidance Platform)

2. REFERENCES

- Software Requirement Specification (SRS) Document

3. INTRODUCTION

Background to the Problem

Students in Bangladesh and similar developing countries often face career confusion because of lack of proper guidance. Many students choose academic paths based on peer pressure, family influence, or limited awareness of market demand. The AI-Based Student Career Guidance Platform project identifies high graduate unemployment, lack of structured career counseling, poor alignment between education and the job market, limited awareness of modern career opportunities such as AI, freelancing and remote jobs, and traditional counseling systems that are manual and outdated. As a result, students may enter careers that do not match their skills or interests.

Existing software solutions such as LinkedIn Learning, Coursera and Udemy mainly support online learning and skill development, while traditional counseling depends heavily on human counselors. The project identifies a gap in localized, personalized career guidance for Bangladeshi students.

Solution to the Problem

The proposed solution is an AI-Based Student Career Guidance Platform that provides personalized and data-driven career recommendations. The system analyzes students' academic background, interests, skills and job-market information to suggest suitable careers, jobs, internships, courses and skill-development opportunities.

The platform is a web-based intelligent career guidance system. Its main purpose is to reduce career confusion, improve employability and bridge the gap between education and industry requirements. Major functions include AI-based career recommendations, skill-gap analysis, resume analysis, career roadmap generation, job/internship/course recommendations, AI chatbot support, administration, notifications and user feedback.

The project proposes React.js, Node.js, MongoDB, Python and AI APIs. The solution is considered technically, economically and operationally feasible according to the original project report.

Testing Focus

- Correctness of each functional module.
- Integration between modules and external services.
- Protection of student and administrator information.
- Clarity and usability of core student workflows.
- Response performance for common actions.

- Reliability and graceful handling of service failures.
- Compatibility across supported browsers and responsive screen sizes.
- Relevance/accuracy of AI-assisted recommendations against predefined test personas and an agreed rubric.

4. REQUIREMENT SPECIFICATION

4.1 System Features

1. User Registration and Authentication

Functional Requirements

- 1.1 The software shall allow a new student to register by providing name, email, password, and other required personal information.
- 1.2 The software shall validate all required registration fields before creating a new account.
- 1.3 The software shall verify whether the provided email address is already registered in the system.
- 1.4 The software shall display an appropriate error message if the email address already exists or if any required information is missing.
- 1.5 The software shall securely create a user account after successful registration.
- 1.6 The software shall allow registered users to log in using their email and password.
- 1.7 The software shall display an error message when invalid login credentials are provided.
- 1.8 The software shall create a user session after successful authentication.
- 1.9 The software shall redirect the authenticated user to the appropriate dashboard or homepage after successful login.

Priority Level: High

Precondition: User must open the registration or login page and provide the required registration or login information.

2. Student Profile Management

Functional Requirements

- 2.1 The software shall allow students to create their personal profile after successful registration.
- 2.2 The software shall allow students to enter their academic background and educational information.
- 2.3 The software shall allow students to provide their interests and career preferences.
- 2.4 The software shall allow students to enter their existing technical and professional skills.
- 2.5 The software shall allow students to view their previously saved profile information.
- 2.6 The software shall allow students to update their personal, academic, interest, and skill information.
- 2.7 The software shall validate required profile information before saving the profile.
- 2.8 The software shall store the student's profile information in the database.

Priority Level: High

Precondition: Student must be registered and logged in to the system.

3. AI-Based Career Recommendation

Functional Requirements

- 3.1 The software shall allow students to request career recommendations based on their profile information.
- 3.2 The software shall analyze the student's academic background, interests, and skills.
- 3.3 The software shall generate suitable career recommendations based on the analyzed information.
- 3.4 The software shall display recommended career options to the student.
- 3.5 The software shall provide relevant information about the recommended career options.
- 3.6 The software shall update career recommendations when relevant student profile information is changed.
- 3.7 The software shall display an appropriate message if insufficient profile information is available for generating recommendations.

Priority Level: High

Precondition: Student must be logged in and provide the required academic, interest, and skill information.

4. Job Recommendation

Functional Requirements

- 4.1 The software shall allow students to view job recommendations based on their skills and career interests.
- 4.2 The software shall analyze the student's profile information to identify relevant job opportunities.
- 4.3 The software shall display suitable job opportunities to the student.
- 4.4 The software shall provide relevant job information such as job title, required skills, and other available details.
- 4.5 The software shall update job recommendations according to the student's updated skills and career preferences.
- 4.6 The software shall display an appropriate message when no relevant job opportunities are available.

Priority Level: High

Precondition: Student must be logged in and have relevant profile, skill, or career information.

5. Internship Recommendation

Functional Requirements

- 5.1 The software shall allow students to view recommended internship opportunities.
- 5.2 The software shall recommend internships based on the student's academic background, skills, and career interests.
- 5.3 The software shall display relevant internship information to the student.
- 5.4 The software shall allow students to review available internship opportunities.
- 5.5 The software shall update internship recommendations according to the student's profile information.

5.6 The software shall display an appropriate message when no suitable internship is available.

Priority Level: High

Precondition: Student must be logged in and provide relevant academic, skill, or career information.

6. Course Recommendation

Functional Requirements

6.1 The software shall allow students to view recommended courses based on their career goals and skill requirements.

6.2 The software shall analyze the student's existing skills to identify suitable learning opportunities.

6.3 The software shall recommend courses that can help students develop required skills.

6.4 The software shall display relevant course information to the student.

6.5 The software shall update course recommendations when the student's career goal or skill information changes.

6.6 The software shall display an appropriate message when no relevant course is available.

Priority Level: Medium

Precondition: Student must be logged in and provide a career goal or relevant skill information.

7. Skill Gap Analysis

Functional Requirements

7.1 The software shall allow students to select a target career for skill gap analysis.

7.2 The software shall compare the student's existing skills with the skills required for the selected career.

7.3 The software shall identify missing or insufficient skills.

7.4 The software shall display the identified skill gaps to the student.

7.5 The software shall provide suggestions for improving the identified skills.

7.6 The software shall update the skill gap analysis when the student's skill information is changed.

Priority Level: High

Precondition: Student must be logged in and provide existing skills and a target career.

8. Resume Upload and Analysis

Functional Requirements

8.1 The software shall allow students to upload their resume.

8.2 The software shall accept supported resume file formats.

8.3 The software shall validate the uploaded resume before processing.

8.4 The software shall analyze the information contained in the uploaded resume.

8.5 The software shall display the resume analysis results to the student.

8.6 The software shall identify relevant skills and information from the uploaded resume.

8.7 The software shall display an appropriate error message when an unsupported or invalid resume is uploaded.

Priority Level: High

Precondition: Student must be logged in and have a valid resume in a supported format.

9. Career Roadmap

Functional Requirements

9.1 The software shall allow students to select a target career.

9.2 The software shall generate a personalized career roadmap based on the student's profile and target career.

9.3 The software shall identify required skills and learning areas for the selected career.

9.4 The software shall display the career roadmap in a structured and understandable manner.

9.5 The software shall provide actionable steps for career development.

9.6 The software shall update the roadmap when relevant student profile or career information changes.

Priority Level: High

Precondition: Student must be logged in and provide a target career and relevant profile information.

10. AI Career Chatbot

Functional Requirements

10.1 The software shall provide an AI chatbot for career-related assistance.

10.2 The software shall allow students to enter career-related questions.

10.3 The software shall process the student's career-related query.

10.4 The software shall provide a relevant response to the student's query.

10.5 The software shall allow students to continue a conversation with the chatbot.

10.6 The software shall display an appropriate message when the chatbot service is temporarily unavailable.

Priority Level: High

Precondition: Student must be logged in and the chatbot service must be available.

11. Admin Dashboard

Functional Requirements

11.1 The software shall provide a dashboard for authorized admin users.

11.2 The software shall allow administrators to access relevant system management functions.

11.3 The software shall allow administrators to monitor user-related information.

11.4 The software shall provide administrative analytics and reporting functions.

11.5 The software shall prevent unauthorized users from accessing the admin dashboard.

11.6 The software shall display an appropriate error or redirect message when an unauthorized user attempts to access admin functions.

Priority Level: High

Precondition: Admin must be logged in with valid administrator credentials.

12. Analytics and Reporting

Functional Requirements

- 12.1** The software shall allow authorized administrators to view system analytics.
- 12.2** The software shall display relevant user and system activity information.
- 12.3** The software shall generate reports based on available system data.
- 12.4** The software shall display analytics information in an understandable format.
- 12.5** The software shall ensure that analytics information is accessible only to authorized administrators.

Priority Level: Medium

Precondition: Admin must be logged in and relevant system data must be available.

13. University Partnership Management

Functional Requirements

- 13.1** The software shall provide a university partnership module.
- 13.2** The software shall allow authorized users to enter relevant university partnership information.
- 13.3** The software shall store valid university partnership information.
- 13.4** The software shall allow authorized users to view available partnership information.
- 13.5** The software shall maintain the accuracy and consistency of partnership information.

Priority Level: Medium

Precondition: Authorized user must be logged in and have access to the university partnership module.

14. Notification System

Functional Requirements

- 14.1** The software shall generate notifications for relevant system events.
- 14.2** The software shall display notifications to the appropriate users.
- 14.3** The software shall notify students about relevant recommendations or system updates.
- 14.4** The software shall display notifications in a clear and understandable format.
- 14.5** The software shall prevent unnecessary duplication of the same notification.

Priority Level: Medium

Precondition: User must be logged in and a notification-triggering event must occur.

15. User Feedback

Functional Requirements

- 15.1** The software shall allow students to submit feedback about the platform.
- 15.2** The software shall provide a feedback form for entering user comments.
- 15.3** The software shall validate required feedback information before submission.
- 15.4** The software shall store submitted feedback in the database.

15.5 The software shall display a confirmation message after successful feedback submission.

15.6 The software shall allow authorized users to review submitted feedback.

Priority Level: Medium

Precondition: Student must be logged in and open the feedback section.

4.2 System Quality Attributes

The AI-Based Student Career Guidance Platform must satisfy several quality attributes to ensure that the system is reliable, secure, usable and effective. These quality attributes are prioritized according to their importance to the system.

1. Accuracy

Priority: High

The system shall provide relevant career, job, internship, course and skill-gap recommendations based on the student's provided academic background, interests, skills and available test data.

2. Personalization

Priority: High

The system shall provide personalized recommendations and career roadmaps that reflect the individual student's academic background, interests, skills and selected career goals.

3. Security and Data Privacy

Priority: High

The system shall protect user authentication information and student data from unauthorized access, modification or inappropriate disclosure.

4. Usability

Priority: High

The system shall provide a simple and understandable interface so that students can complete common tasks without unnecessary steps or confusing navigation.

5. Performance

Priority: High

The system shall respond to normal user actions and AI-supported operations within the agreed response-time threshold under the defined testing environment.

6. Reliability

Priority: High

The system shall perform its core functions consistently without unexpected crashes, failures or loss of important user data during repeated testing.

7. Availability

Priority: Medium

The system shall remain available during planned testing, demonstration and acceptance-testing periods.

8. Data Integrity

Priority: High

The system shall maintain the correctness and consistency of student profiles, resumes, feedback, recommendations and administrative data after create and update operations.

9. Compatibility

Priority: Medium

The web-based system shall function correctly on supported modern web browsers and common desktop and mobile screen sizes.

10. Scalability

Priority: Medium

The system architecture shall support an increase in users and system data without unacceptable performance degradation within the target scope of the project.

11. Responsiveness

Priority: High

The AI chatbot and notification interfaces shall respond appropriately to user actions and provide clear status or confirmation feedback.

4.3 System Interface

The system interface is a web-based interface through which students, administrators and supporting stakeholders interact with the platform.

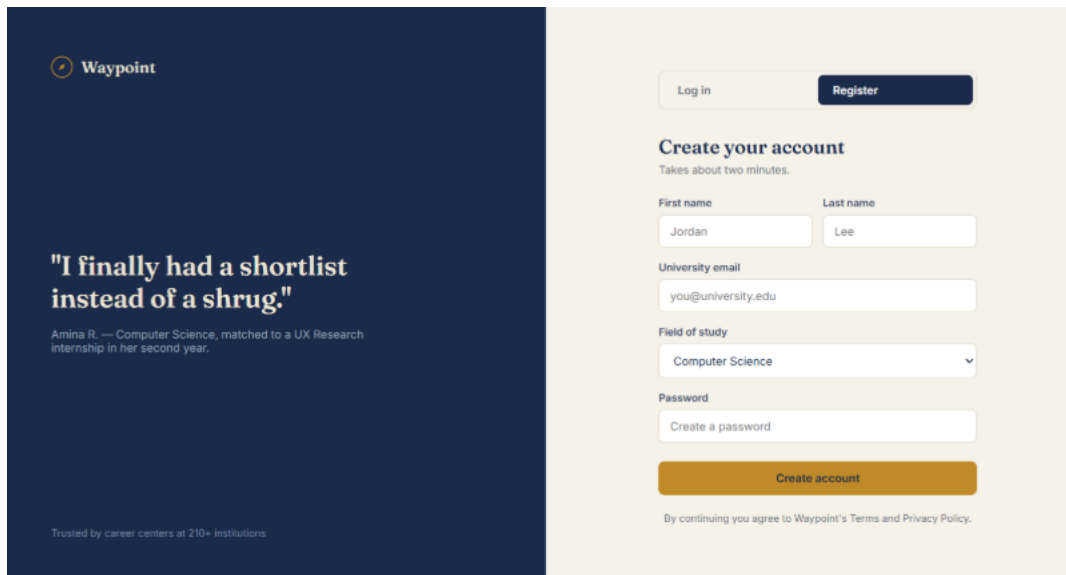
1. Login Page

Students and administrators can securely log in to their Waypoint accounts using their registered credentials. The page provides a simple and user-friendly authentication experience.

Fig 1: Login page

2. Signup Page

New users can create a Waypoint account by providing their basic information and account credentials. The registration process is designed to be quick, simple, and secure.



The image shows the Waypoint registration page. On the left, a dark blue sidebar contains the Waypoint logo and a testimonial from Amina R. On the right, a light beige form titled 'Create your account' includes fields for first and last names, university email, field of study, and password. It also features 'Log in' and 'Register' buttons at the top and a 'Create account' button at the bottom.

Waypoint

"I finally had a shortlist instead of a shrug."

Amina R. — Computer Science, matched to a UX Research internship in her second year.

Trusted by career centers at 210+ institutions

Log in Register

Create your account
Takes about two minutes.

First name Last name
Jordan Lee

University email
you@university.edu

Field of study
Computer Science

Password
Create a password

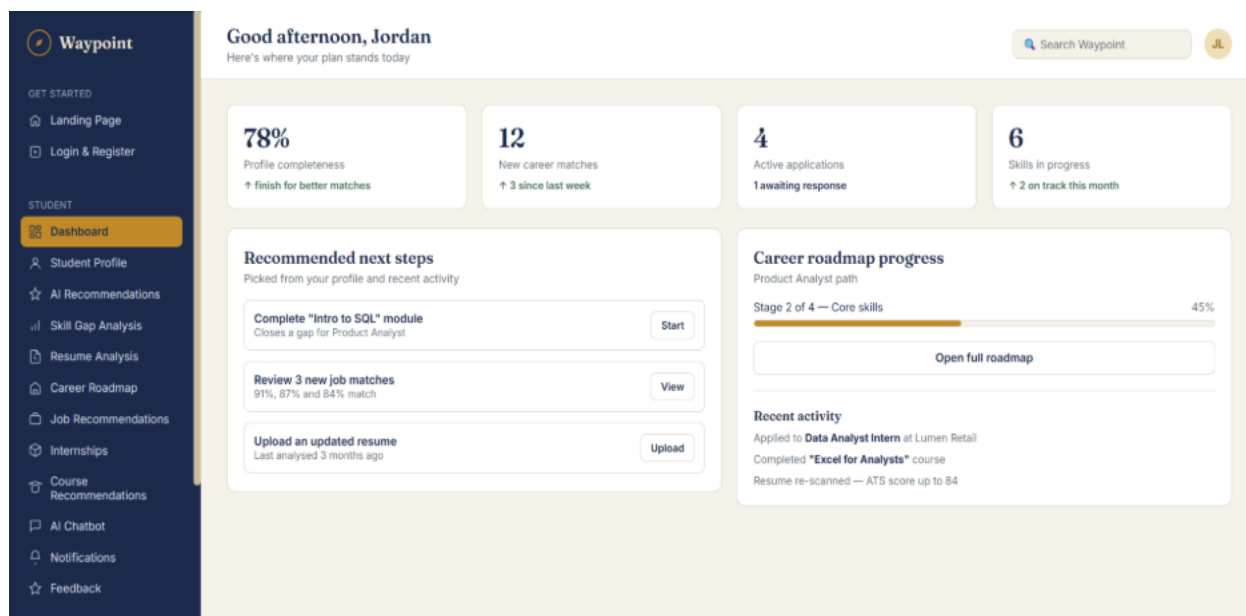
Create account

By continuing you agree to Waypoint's Terms and Privacy Policy.

Fig 2: Signup Page

3. User Dashboard Page

The dashboard provides students with an overview of their career journey, recommendations, activities, and notifications. Users can easily navigate to all major features from one place.



The image shows the Waypoint user dashboard for a student named Jordan. It features a dark blue sidebar with navigation links, a top header with a greeting and search bar, and a main content area with four summary cards and two detailed sections: 'Recommended next steps' and 'Career roadmap progress'.

Waypoint

GET STARTED
Landing Page
Login & Register

STUDENT
Dashboard
Student Profile
AI Recommendations
Skill Gap Analysis
Resume Analysis
Career Roadmap
Job Recommendations
Internships
Course Recommendations
AI Chatbot
Notifications
Feedback

Good afternoon, Jordan
Here's where your plan stands today

Search Waypoint JL

78%
Profile completeness
↑ finish for better matches

12
New career matches
↑ 3 since last week

4
Active applications
1 awaiting response

6
Skills in progress
↑ 2 on track this month

Recommended next steps
Picked from your profile and recent activity

Complete "Intro to SQL" module
Start
Closes a gap for Product Analyst

Review 3 new job matches
View
91%, 87% and 84% match

Upload an updated resume
Upload
Last analysed 3 months ago

Career roadmap progress
Product Analyst path

Stage 2 of 4 — Core skills 45%

Open full roadmap

Recent activity
Applied to Data Analyst Intern at Lumen Retail
Completed "Excel for Analysts" course
Resume re-scanned — ATS score up to 84

Fig 3: User Dashboard Page

4. Admin Dashboard Page

The admin dashboard allows administrators to manage users, monitor platform activities, and maintain career-related content. It provides centralized control over the Waypoint platform.

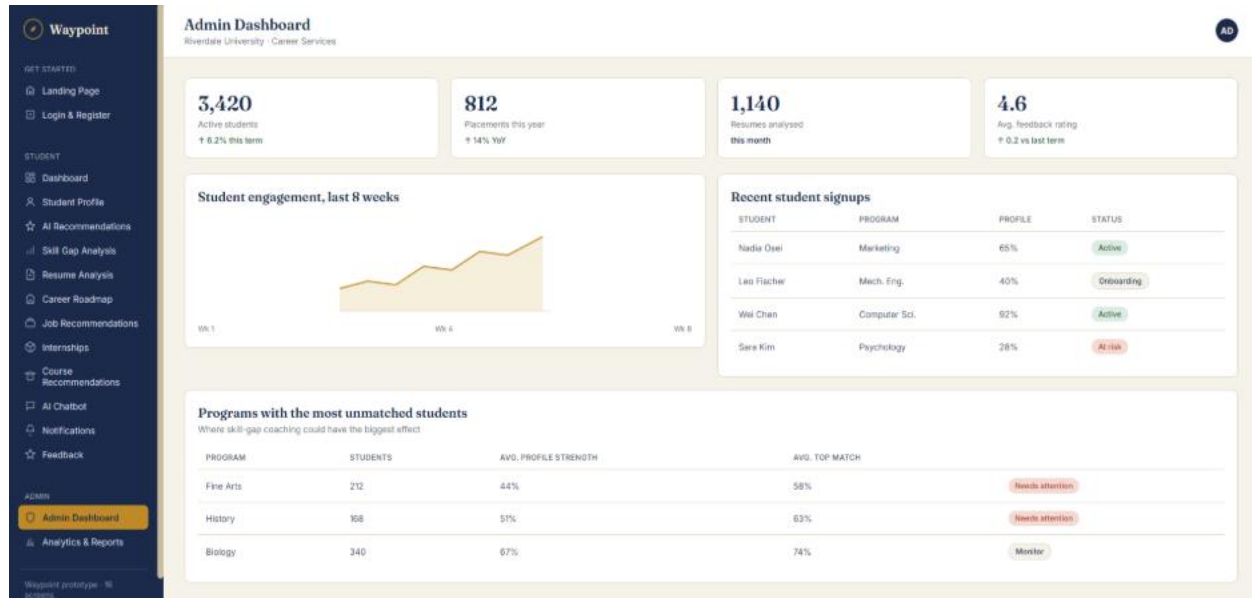


Fig 4: Admin Dashboard Page

5. Student Profile Page

Students can create and manage their personal, academic, technical, and career-related information. This profile data helps Waypoint generate more personalized recommendations.

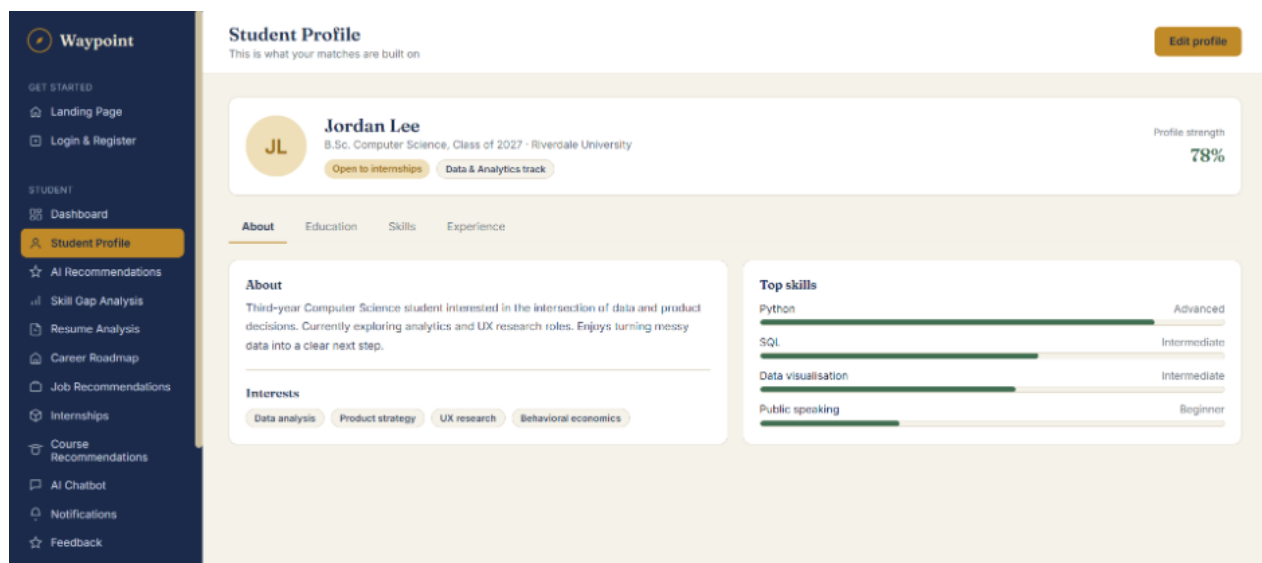


Fig 5: Student Profile Page

6. AI Career Recommendations Page

This page uses AI to analyze the student's skills, interests, academic background, and goals. Based on the analysis, Waypoint suggests suitable career paths and job roles.

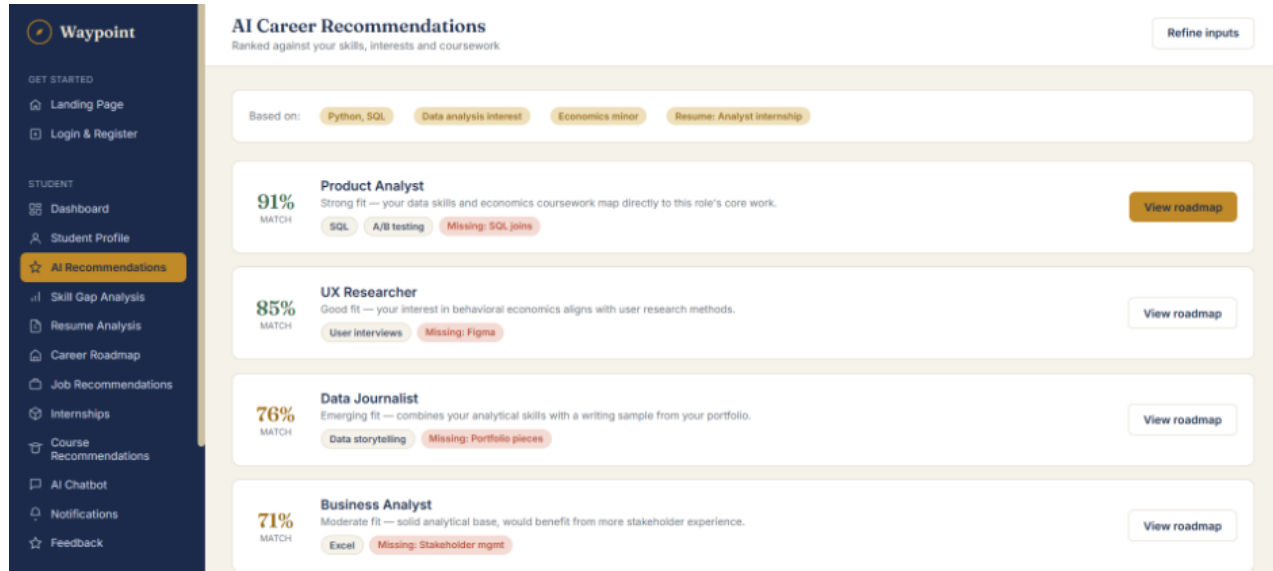


Fig 6: AI Career Recommendations Page

7. Resume Upload & Analysis Page

Students can upload their resumes for AI-powered analysis. Waypoint identifies strengths, weaknesses, missing skills, and improvement areas to make the resume more career-ready.

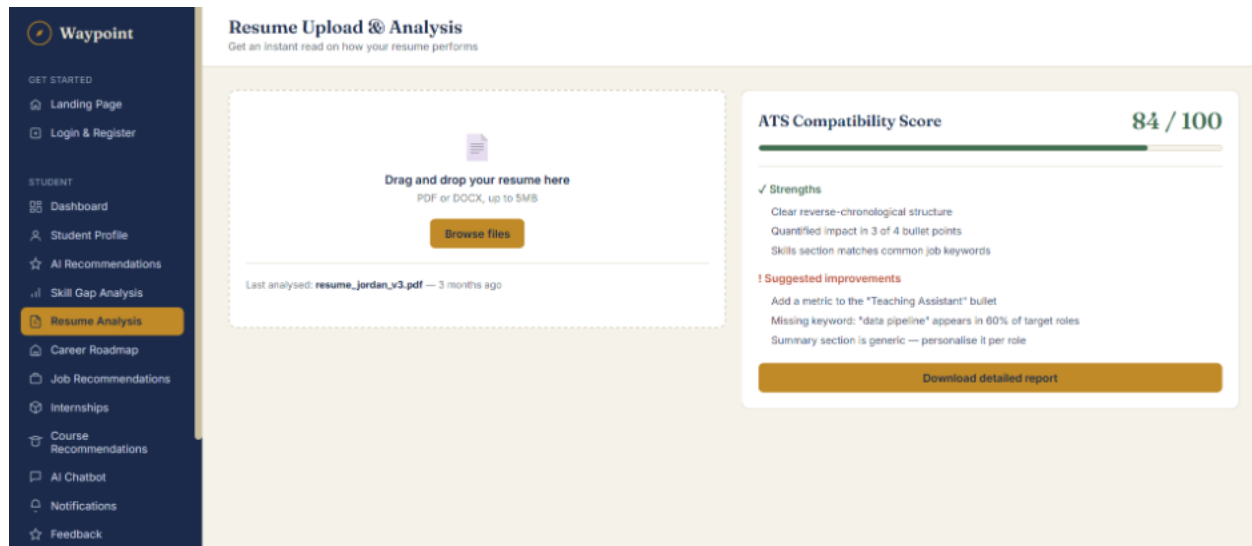


Fig 7: Resume Upload & Analysis Page

8. Career Roadmap Page

The Career Roadmap provides students with a personalized step-by-step career development plan. It highlights the skills, courses, projects, certifications, and milestones needed to achieve their career goals.

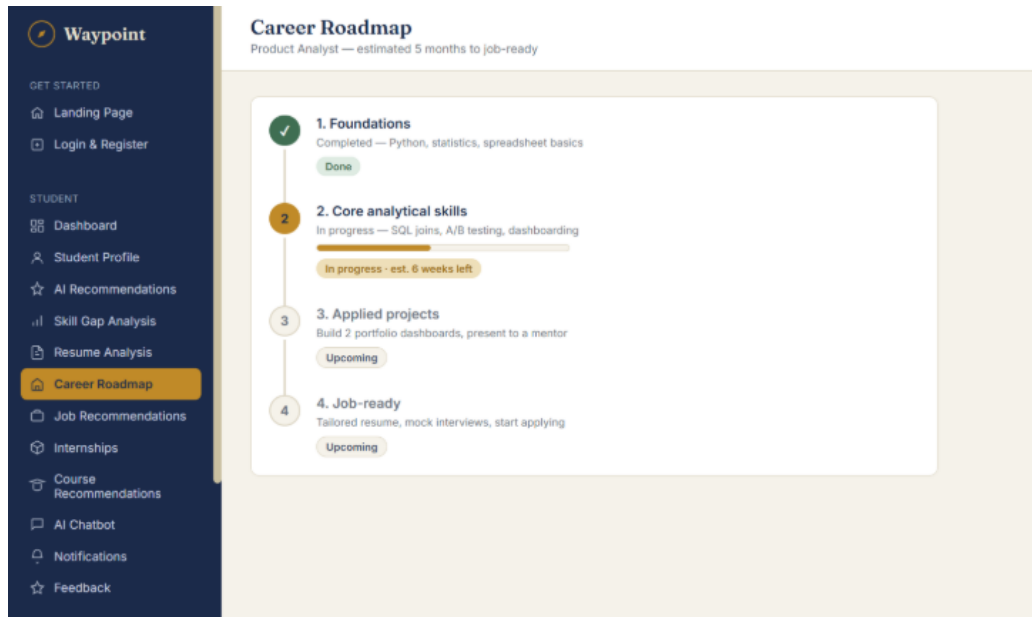


Fig 8: Career Roadmap Page

9. Job Recommendations Page

Students receive personalized job recommendations based on their skills, qualifications, interests, and career preferences. The page helps users discover relevant employment opportunities.

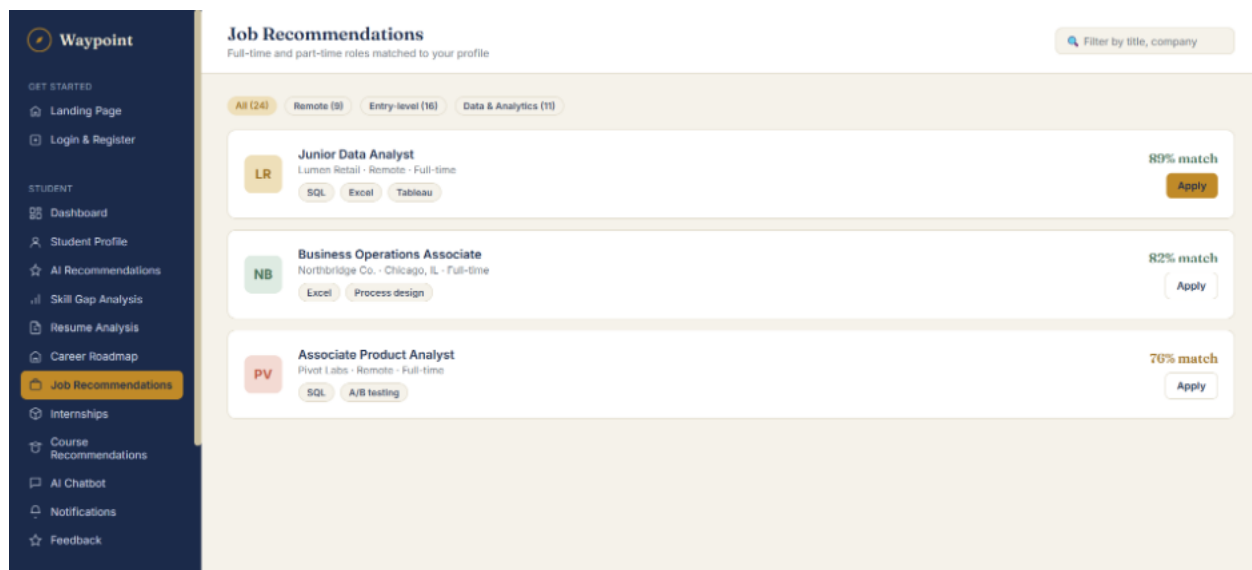


Fig 9: Job Recommendations Page

10. Internship Recommendations Page

This page recommends suitable internship opportunities for students based on their academic background and skills. It helps students gain relevant industry experience and prepare for future careers.

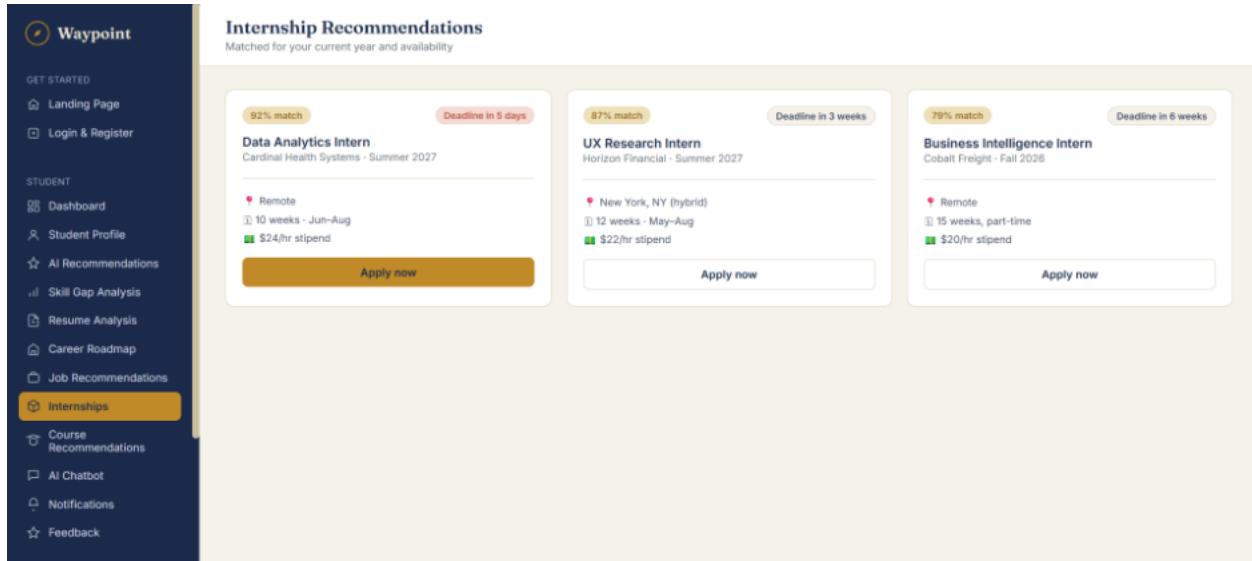


Fig 10: Internship Recommendations Page

11. Course Recommendations Page

Waypoint suggests relevant online and professional courses based on the student's skill gaps and career goals. These recommendations help students develop the knowledge and skills required for their desired career.

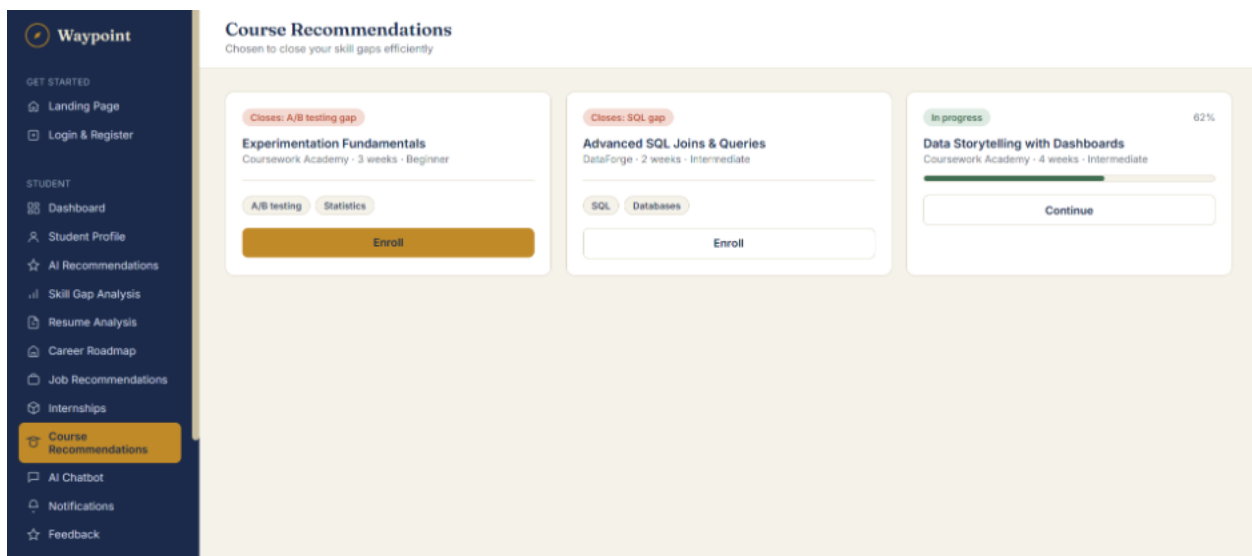


Fig 11: Course Recommendations Page

12. Ask Waypoint Page

Students can interact with the Waypoint AI assistant to ask career-related questions. The AI provides personalized guidance on careers, skills, resumes, courses, jobs, and internships.

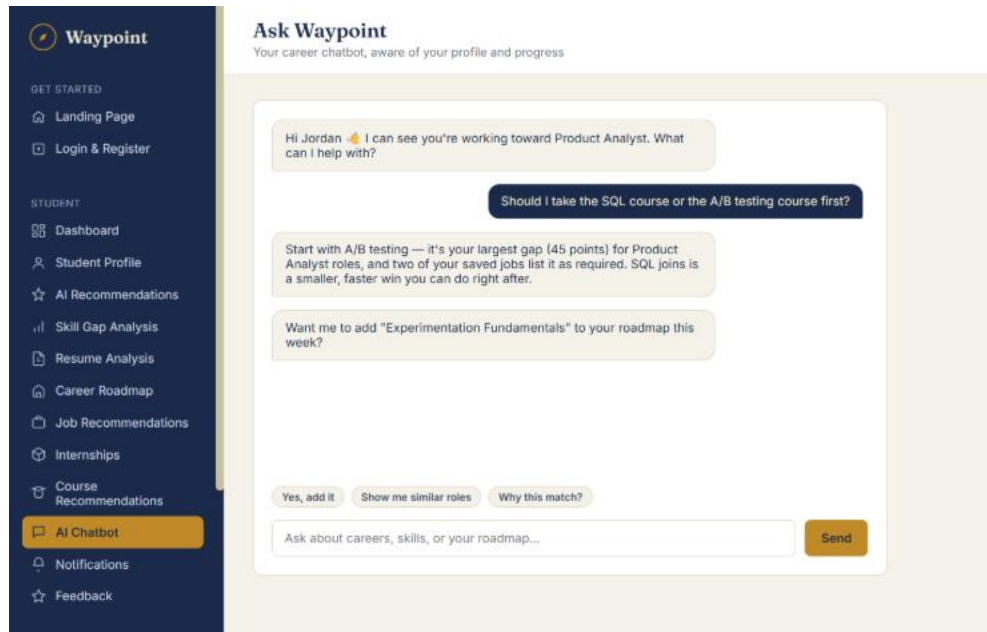


Fig 12: Ask Waypoint Page

13. Notifications Page

The Notifications page keeps users updated about new job and internship opportunities, course recommendations, roadmap milestones, and important platform activities.

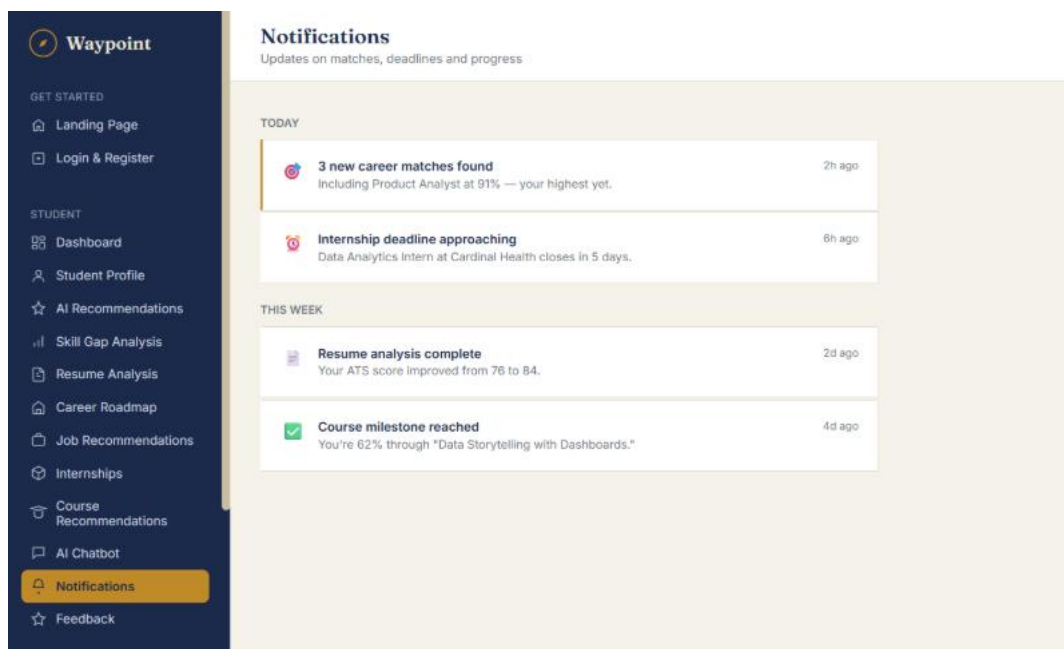


Fig 13: Notifications Page

14. Feedback Page

Students can share their opinions, suggestions, and experiences regarding the Waypoint platform. The collected feedback helps administrators improve the system and user experience.

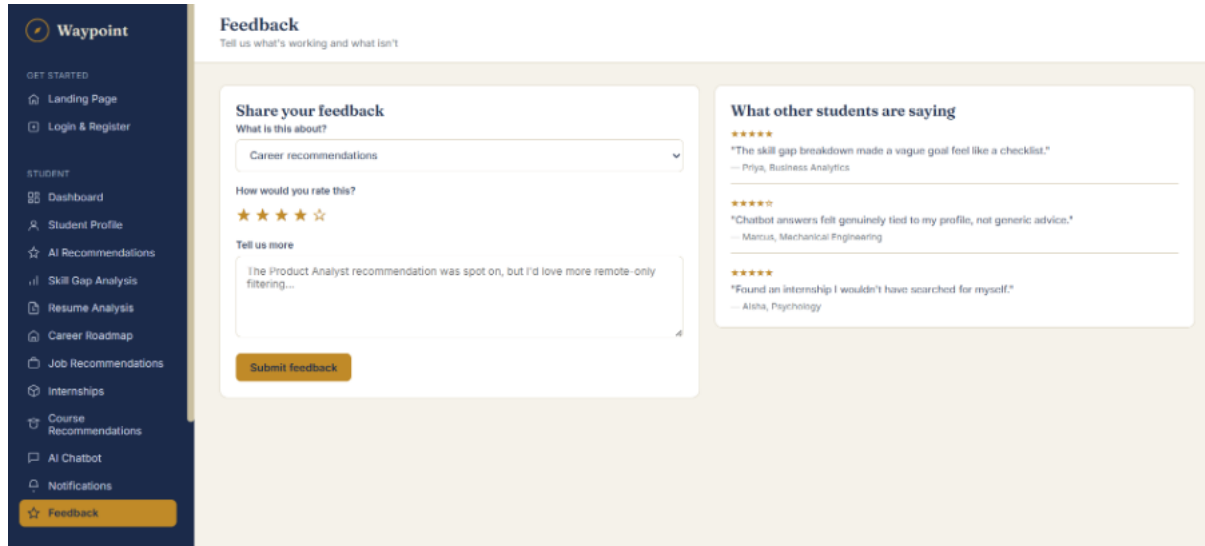


Fig 14: Feedback Page

4.4 Project Requirements

The Constructive Cost Model (COCOMO) is an algorithmic software cost estimation model used to estimate the effort, development time, and required team size of a software project. For the AI-Based Student Career Guidance Platform, the semi-detached model is appropriate because the project involves moderate complexity, AI-based functionality, database integration, web development, and a small development team with moderate experience.

Using the semi-detached model coefficients:

Coefficient (a) = 3.0

Effort Exponent (b) = 1.12

Time Exponent (c) = 0.35

The AI-Based Student Career Guidance Platform consists of approximately 6,000 Source Lines of Code (SLOC) based on the estimated frontend, backend, database integration, AI-related components, authentication, recommendation modules, and administrative functionalities.

Effort Estimation

The COCOMO effort equation is:

$$\text{Effort (PM)} = 3.0 \times (\text{KLOC})^{1.12}$$

Here,

$$\text{KLOC} = 6,000 / 1,000 = 6 \text{ KLOC}$$

$$\text{Effort} = 3.0 \times (6)^{1.12}$$

$$= 3.0 \times 7.66$$

$$= 22.98 \text{ person-months}$$

Therefore, the estimated development effort is approximately:

$$\text{Effort} \approx 23 \text{ person-months}$$

Development Time Estimation

The development time is calculated using:

$$\text{Development Time} = 2.5 \times (\text{Effort})^{0.35}$$

$$= 2.5 \times 2.99$$

$$= 7.48 \text{ months}$$

Development Time $\approx$ 7.5 months or approximately 8 months.

Required Number of People

The average team size is calculated as:

$$\text{Team Size} = \text{Effort} / \text{Development Time}$$

$$= 22.98 / 7.48$$

$$\approx 3.07$$

Therefore, approximately 3 people are required for the project.

Cost Estimation

Assuming the average monthly cost per developer is 30,000 BDT:

$$\text{Developer Salary} = 3 \times 30,000 \times 8$$

$$= 720,000 \text{ BDT}$$

Additional project expenses are estimated as:

$$\text{Software/Cloud/API Cost} = 100,000 \text{ BDT}$$

$$\text{Testing and Deployment Cost} = 50,000 \text{ BDT}$$

$$\text{Transport and Communication Cost} = 40,000 \text{ BDT}$$

$$\text{Miscellaneous Expenses} = 40,000 \text{ BDT}$$

Therefore,

$$\text{Total Project Budget} = 720,000 + 100,000 + 50,000 + 40,000 + 40,000$$

$$= 950,000 \text{ BDT}$$

5. FEATURES NOT TO BE TESTED

- Internal implementation of third-party AI models or APIs; only observable integration behavior and returned output are tested.
- Internal algorithms or infrastructure of external job, internship and course providers.
- Production-scale capacity beyond the available student-project test environment.
- Real financial/payment transactions because no current SRE functional requirement specifies a payment workflow.
- Formal legal/compliance certification and professional counselor certification.
- Internal systems of university partners that are outside the project team's control.
- Internal training process of third-party AI models unless directly implemented as part of this project.

6. TESTING APPROACH

Testing will be performed using a requirements-driven and risk-based approach. High-priority requirements receive greater testing attention. Unit, System/Integration and Acceptance testing are the primary levels, following the structure of the official template. Regression testing will be performed after major fixes.

6.1 Testing Levels

The testing for the AI-Based Student Career Guidance Platform will involve multiple levels, including Unit Testing, Integration Testing, System Testing, and User Acceptance Testing (UAT). Since the system contains several interconnected modules such as user authentication, student profile management, AI-based career recommendation, job and internship recommendation, course recommendation, skill gap analysis, resume analysis, career roadmap, AI chatbot, notification, feedback, and admin management, a comprehensive testing approach will be followed to ensure that the system is reliable, secure, accurate, and user-friendly. The development team and QA/test personnel will participate throughout the testing lifecycle.

• Unit Testing

Unit testing will be conducted by the developers for individual components and functions of the AI-Based Student Career Guidance Platform. The main objective is to verify that each module works correctly before it is integrated with other modules.

The unit testing process will focus on individual functionalities such as user registration and login, student profile management, career recommendation, job recommendation, internship suggestion, course recommendation, skill gap analysis, resume processing, career roadmap generation, AI chatbot, notification handling, feedback submission, and administrative functions.

Each developer will prepare and execute appropriate test cases to verify expected system behavior, validate input fields, handle invalid inputs, and verify database interactions. For AI-based modules, the developer will also verify whether valid profile information is correctly processed and whether appropriate recommendations or analysis results are generated.

Unit testing results, including test cases, sample outputs, and identified defects, will be documented and provided to the QA/test lead for further evaluation. A module will proceed to integration testing after the critical defects identified during unit testing have been corrected.

- **Integration Testing**

Integration testing will focus on verifying the interaction between different modules after they are combined. The main objective is to ensure that data is transferred correctly between interconnected components and that one module does not negatively affect another module.

The integration testing will include the interaction between user registration and authentication, student profile management and AI career recommendation, profile information and skill gap analysis, resume upload and resume analysis, career selection and career roadmap generation, profile information and job/internship/course recommendations, and chatbot services.

The integration between the application and the MongoDB database will also be tested to ensure that student profiles, resume information, recommendations, feedback, notifications, and administrative information are stored and retrieved correctly.

Integration testing will also verify authentication sessions, authorization, data flow between frontend and backend, API communication, and interactions with AI services. The development team and QA/test lead will jointly perform these tests. Any integration defects identified during this phase will be documented, corrected, and retested.

- **System Testing**

System testing will evaluate the complete AI-Based Student Career Guidance Platform as an integrated system to determine whether it satisfies the defined functional and non-functional requirements.

The testing will cover end-to-end workflows of the platform, including student registration and login, profile creation and update, AI-based career recommendation, job and internship recommendation, course recommendation, skill gap analysis, resume upload and analysis, career roadmap generation, AI chatbot interaction, notifications, feedback submission, and administrative operations.

The frontend interface will be tested for navigation, forms, input validation, displayed information, responsiveness, and usability. Backend functionality will be tested for authentication, authorization, database operations, API communication, recommendation processing, and data integrity.

Special attention will be given to the accuracy and consistency of AI-supported functions. Different student profiles will be used to verify whether the system provides relevant recommendations according to the provided academic background, interests, skills, and career goals.

System testing will also evaluate security, data privacy, reliability, browser compatibility, and basic performance under normal user activities. The objective is to ensure that the complete system works consistently and provides an acceptable experience before it proceeds to acceptance testing.

- **User Acceptance Testing (UAT)**

User Acceptance Testing will involve actual or representative student users with assistance from the QA/test lead and development team. The purpose of UAT is to determine whether the completed system is suitable for its intended users and whether it satisfies the practical requirements of student career guidance. During UAT, users will perform representative activities such as creating an account, logging in, completing their student profile, receiving career recommendations, checking recommended jobs and internships, viewing recommended courses, performing skill gap analysis, uploading and analyzing a resume, generating a career roadmap, interacting with the AI chatbot, viewing notifications, and submitting feedback.

Users will also evaluate the overall usability of the platform, including navigation, clarity of information, ease of completing tasks, usefulness of recommendations, and responsiveness of the system.

Feedback collected from users during UAT will be documented and analyzed. Any important usability or functional issues identified during this phase will be reported to the development team for correction and retesting.

The acceptance testing will be considered successful when the representative users can complete the major career-guidance workflows without critical problems and the system satisfies the agreed acceptance criteria.

6.2 Test Tools

This table presents the tools selected for software testing and their respective purposes. It covers API, browser, performance, database, automation, and project management activities.

Tool	Purpose
Postman	API functional and integration testing.
Browser Developer Tools	Console, network, storage, responsive and client-side issue investigation.
JMeter or equivalent	Basic performance/load testing where feasible.
Git/GitHub	Version control and change/fix traceability.
MongoDB tools	Database verification and data integrity checking.
Selenium/Playwright (if available)	Repeatable browser regression testing.
Asana	Project testing schedule/timeline; required project-management tool for the schedule.

7. TEST CASES/TEST ITEMS

Test Case ID: FR_1

Verify the system provides a suitable career recommendation from student academic background, interests and skills.

Project Name: AI-Based Student Career Guidance Platform			Test Designed by: Wafa		
Test Case ID: FR_1			Test Designed date: 31-08-2026		
Test Priority (Low, Medium, High): High			Test Executed by: Muhammad Talha Islam		
Module Name: AI Career Engine			Test Execution date: 15-11-2026		
Test Title: Verify career recommendation with a valid student profile					
Description: Verify the system provides a suitable career recommendation from student academic background, interests and skills.					
Precondition (If any): Student is registered and logged in; profile contains required information.					
Test Steps	Test Data	Expected Results	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as student 2. Open/complete profile 3. Enter academic background, interests and skills 4. Open Career Recommendation 5. Submit request	Student profile: CSE; interests: AI/software; skills: Python, SQL, JavaScript.	System analyzes the supplied profile and displays suitable career recommendations.	System analyzes the supplied profile and displays suitable career recommendation s.	To be completed during execution.	Not Executed
Post Condition: System remains stable and test data is retained or cleared according to the test-data policy.					

Test Case ID: FR_2

Verify the system recommends relevant jobs based on available profile information.

Project Name: AI-Based Student Career Guidance Platform			Test Designed by: Wafa		
Test Case ID: FR_2			Test Designed date: 31-08-2026		
Test Priority (Low, Medium, High): High			Test Executed by: Muhammad Talha Islam		
Module Name: AI Career Engine			Test Execution date: 15-11-2026		
Test Title: Verify job recommendation					
Description: Verify the system recommends relevant jobs based on available profile information.					
Precondition (If any): Student is authenticated and profile data is available.					
Test Steps	Test Data	Expected Results	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Job Recommendation 2. Request recommendations 3. Review returned jobs	Profile with Python, SQL and web-development skills; controlled job records.	Relevant job recommendations are displayed without application error.	Relevant job recommendations are displayed without application error.	To be completed during execution.	Not Executed
Post Condition: System remains stable and test data is retained or cleared according to the test-data policy.					

Test Case ID: FR_3

Verify the system suggests relevant internship opportunities.

Project Name: AI-Based Student Career Guidance Platform			Test Designed by: Wafa		
Test Case ID: FR_3			Test Designed date: 31-08-2026		
Test Priority (Low, Medium, High): High			Test Executed by: Muhammad Talha Islam		
Module Name: AI Career Engine			Test Execution date: 15-11-2026		
Test Title: Verify internship suggestion					
Description: Verify the system suggests relevant internship opportunities.					
Precondition (If any): Student is authenticated and profile data is available.					
Test Steps	Test Data	Expected Results	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Internship Suggestions 2. Request suggestions 3. Review result list	Student profile and controlled internship records.	Relevant internship opportunities are displayed.	Relevant internship opportunities are displayed.	To be completed during execution.	Not Executed
Post Condition: System remains stable and test data is retained or cleared according to the test-data policy.					

Test Case ID: FR_4

Verify the system recommends courses supporting career or skill development.

Project Name: AI-Based Student Career Guidance Platform			Test Designed by: Wafa		
Test Case ID: FR_4			Test Designed date: 31-08-2026		
Test Priority (Low, Medium, High): Medium			Test Executed by: Muhammad Talha Islam		
Module Name: AI Career Engine			Test Execution date: 15-11-2026		
Test Title: Verify course recommendation					
Description: Verify the system recommends courses supporting career or skill development.					
Precondition (If any): Student profile contains a target interest/skill area.					
Test Steps	Test Data	Expected Results	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Course Recommendation 2. Request recommendations 3. Review course list	Target skill: Python/AI; controlled course records.	Suitable courses are displayed for the student's career/skill needs.	Suitable courses are displayed for the student's career/skill needs.	To be completed during execution.	Not Executed
Post Condition: System remains stable and test data is retained or cleared according to the test-data policy.					

Test Case ID: FR_5

Verify a new user can register and then authenticate with valid credentials.

Project Name: AI-Based Student Career Guidance Platform			Test Designed by: Wafa		
Test Case ID: FR_5			Test Designed date: 31-08-2026		
Test Priority (Low, Medium, High): High			Test Executed by: Muhammad Talha Islam		
Module Name: User Management			Test Execution date: 15-11-2026		
Test Title: Verify registration and valid authentication					
Description: Verify a new user can register and then authenticate with valid credentials.					
Precondition (If any): Registration page is available and test account does not already exist.					
Test Steps	Test Data	Expected Results	Expected Results	Actual Results	Status (Pass/Fail)
1. Open registration 2. Enter valid registration data 3. Submit 4. Open login 5. Enter valid credentials 6. Submit	Name: Test Student; Email: student01@test.local; Password: Valid@123.	Registration succeeds and valid credentials allow secure login.	Registration succeeds and valid credentials allow secure login.	To be completed during execution.	Not Executed
Post Condition: System remains stable and test data is retained or cleared according to the test-data policy.					

Test Case ID: FR_6

Verify a student can create and maintain profile information.

Project Name: AI-Based Student Career Guidance Platform			Test Designed by: Muhammad Talha Islam		
Test Case ID: FR_6			Test Designed date: 31-08-2026		
Test Priority (Low, Medium, High): High			Test Executed by: Wafa		
Module Name: User Management			Test Execution date: 15-11-2026		
Test Title: Verify student profile creation and update					
Description: Verify a student can create and maintain profile information.					
Precondition (If any): Student is authenticated.					
Test Steps	Test Data	Expected Results	Expected Results	Actual Results	Status (Pass/Fail)
1. Open profile 2. Enter academic/interests/skills 3. Save 4. Reopen profile 5. Change one value 6. Save again	Academic: CSE; Interest: AI; Skills: Python, SQL.	Profile is saved and updated information persists correctly.	Profile is saved and updated information persists correctly.	To be completed during execution.	Not Executed
Post Condition: System remains stable and test data is retained or cleared according to the test-data policy.					

Test Case ID: FR_7

Verify the system identifies skill gaps for a selected career.

Project Name: AI-Based Student Career Guidance Platform			Test Designed by: Muhammad Talha Islam		
Test Case ID: FR_7			Test Designed date: 31-08-2026		
Test Priority (Low, Medium, High): High			Test Executed by: Wafa		
Module Name: Skill Analysis			Test Execution date: 15-11-2026		
Test Title: Verify skill gap analysis					
Description: Verify the system identifies skill gaps for a selected career.					
Precondition (If any): Student profile and target career are available.					
Test Steps	Test Data	Expected Results	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Skill Gap Analysis 2. Select target career 3. Generate analysis 4. Review gaps	Skills: Python, SQL; Target: AI/ML career.	Relevant skill gaps and improvement suggestions are displayed.	Relevant skill gaps and improvement suggestions are displayed.	To be completed during execution.	Not Executed
Post Condition: System remains stable and test data is retained or cleared according to the test-data policy.					

Test Case ID: FR_8

Verify the system accepts a supported resume and provides analysis.

Project Name: AI-Based Student Career Guidance Platform			Test Designed by: Muhammad Talha Islam		
Test Case ID: FR_8			Test Designed date: 31-08-2026		
Test Priority (Low, Medium, High): High			Test Executed by: Wafa		
Module Name: Resume Analyzer			Test Execution date: 15-11-2026		
Test Title: Verify valid resume upload and analysis					
Description: Verify the system accepts a supported resume and provides analysis.					
Precondition (If any): Student is authenticated; supported resume is available.					
Test Steps	Test Data	Expected Results	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Resume Analyzer 2. Choose supported resume 3. Upload 4. Start analysis 5. Review result	Representative PDF resume.	Resume is accepted and analysis is displayed.	Resume is accepted and analysis is displayed.	To be completed during execution.	Not Executed
Post Condition: System remains stable and test data is retained or cleared according to the test-data policy.					

Test Case ID: FR_9

Verify the system generates a career roadmap for a student's selected career.

Project Name: AI-Based Student Career Guidance Platform			Test Designed by: Wafa		
Test Case ID: FR_9			Test Designed date: 31-08-2026		
Test Priority (Low, Medium, High): High			Test Executed by: Muhammad Talha Islam		
Module Name: Career Roadmap			Test Execution date: 15-11-2026		
Test Title: Verify personalized career roadmap generation					
Description: Verify the system generates a career roadmap for a student's selected career.					
Precondition (If any): Student profile and target career are available.					
Test Steps	Test Data	Expected Results	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Career Roadmap 2. Select target career 3. Generate roadmap 4. Review steps	Target career: AI/ML Engineer; student skill profile.	A clear, useful and personalized roadmap is generated.	A clear, useful and personalized roadmap is generated.	To be completed during execution.	Not Executed
Post Condition: System remains stable and test data is retained or cleared according to the test-data policy.					

Test Case ID: FR_10

Verify the chatbot responds to a valid career-related question.

Project Name: AI-Based Student Career Guidance Platform			Test Designed by: Wafa		
Test Case ID: FR_10			Test Designed date: 31-08-2026		
Test Priority (Low, Medium, High): High			Test Executed by: Muhammad Talha Islam		
Module Name: AI Chatbot			Test Execution date: 15-11-2026		
Test Title: Verify chatbot answers career-related query					
Description: Verify the chatbot responds to a valid career-related question.					
Precondition (If any): Student is authenticated and chatbot service is available.					
Test Steps	Test Data	Expected Results	Expected Results	Actual Results	Status (Pass/Fail)
1. Open chatbot 2. Enter career-related question 3. Submit 4. Review response	Question: What skills should I learn for an entry-level AI career?	Chatbot returns a relevant and readable career-related response.	Chatbot returns a relevant and readable career-related response.	To be completed during execution.	Not Executed
Post Condition: System remains stable and test data is retained or cleared according to the test-data policy.					

Test Case ID: FR_11

Verify an authorized administrator can access administrative dashboard functions.

Project Name: AI-Based Student Career Guidance Platform			Test Designed by: Wafa		
Test Case ID: FR_11			Test Designed date: 31-08-2026		
Test Priority (Low, Medium, High): High			Test Executed by: Muhammad Talha Islam		
Module Name: Admin Management			Test Execution date: 15-11-2026		
Test Title: Verify authorized admin dashboard access					
Description: Verify an authorized administrator can access administrative dashboard functions.					
Precondition (If any): Valid admin account exists.					
Test Steps	Test Data	Expected Results	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as admin 2. Open dashboard 3. Review administrative functions	Valid admin credentials.	Authorized admin can access the dashboard and permitted functions.	Authorized admin can access the dashboard and permitted functions.	To be completed during execution.	Not Executed
Post Condition: System remains stable and test data is retained or cleared according to the test-data policy.					

Test Case ID: FR_12

Verify authorized administrators can view analytics/reporting information.

Project Name: AI-Based Student Career Guidance Platform			Test Designed by: Muhammad Talha Islam		
Test Case ID: FR_12			Test Designed date: 31-08-2026		
Test Priority (Low, Medium, High): Medium			Test Executed by: Wafa		
Module Name: Admin Management			Test Execution date: 15-11-2026		
Test Title: Verify analytics reporting					
Description: Verify authorized administrators can view analytics/reporting information.					
Precondition (If any): Admin is authenticated; controlled data exists.					
Test Steps	Test Data	Expected Results	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Analytics 2. Generate/view report 3. Compare visible values with controlled test data	Known test records and expected totals.	Analytics are displayed and correspond to the controlled test data.	Analytics are displayed and correspond to the controlled test data.	To be completed during execution.	Not Executed
Post Condition: System remains stable and test data is retained or cleared according to the test-data policy.					

Test Case ID: FR_13

Verify the partnership module handles valid university partnership information.

Project Name: AI-Based Student Career Guidance Platform			Test Designed by: Muhammad Talha Islam		
Test Case ID: FR_13			Test Designed date: 31-08-2026		
Test Priority (Low, Medium, High): Medium			Test Executed by: Wafa		
Module Name: University Partnership			Test Execution date: 15-11-2026		
Test Title: Verify university partnership function					
Description: Verify the partnership module handles valid university partnership information.					
Precondition (If any): Authorized user has access to the module.					
Test Steps	Test Data	Expected Results	Expected Results	Actual Results	Status (Pass/Fail)
1. Open partnership module 2. Enter valid partnership information 3. Save 4. View saved information	Sample university name, contact and partnership details.	Valid partnership information is accepted and displayed correctly.	Valid partnership information is accepted and displayed correctly.	To be completed during execution.	Not Executed
Post Condition: System remains stable and test data is retained or cleared according to the test-data policy.					

Test Case ID: FR_14

Verify a relevant event results in an appropriate notification.

Project Name: AI-Based Student Career Guidance Platform			Test Designed by: Wafa		
Test Case ID: FR_14			Test Designed date: 31-08-2026		
Test Priority (Low, Medium, High): Medium			Test Executed by: Muhammad Talha Islam		
Module Name: Notification System			Test Execution date: 15-11-2026		
Test Title: Verify notification generation and display					
Description: Verify a relevant event results in an appropriate notification.					
Precondition (If any): Notification-triggering event is configured.					
Test Steps	Test Data	Expected Results	Expected Results	Actual Results	Status (Pass/Fail)
1. Trigger defined event 2. Open notification area 3. Review notification	Controlled event such as a new recommendation/update.	Relevant notification is generated and displayed without unexpected duplication.	Relevant notification is generated and displayed without unexpected duplication.	To be completed during execution.	Not Executed
Post Condition: System remains stable and test data is retained or cleared according to the test-data policy.					

Test Case ID: FR_15

Verify an authenticated user can submit feedback.

Project Name: AI-Based Student Career Guidance Platform			Test Designed by: Wafa		
Test Case ID: FR_15			Test Designed date: 31-08-2026		
Test Priority (Low, Medium, High): Medium			Test Executed by: Muhammad Talha Islam		
Module Name: Feedback System			Test Execution date: 15-11-2026		
Test Title: Verify user feedback submission					
Description: Verify an authenticated user can submit feedback.					
Precondition (If any): Student is authenticated and feedback page is available.					
Test Steps	Test Data	Expected Results	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Feedback 2. Enter feedback 3. Submit 4. Review confirmation 5. Check authorized feedback view	Feedback: The career recommendation page is easy to understand.	Feedback is accepted, confirmation is displayed and the authorized recipient can access the record.	Feedback is accepted, confirmation is displayed and the authorized recipient can access the record.	To be completed during execution.	Not Executed
Post Condition: System remains stable and test data is retained or cleared according to the test-data policy.					

8. ITEM PASS/FAIL CRITERIA

A test item will be marked Pass when the actual result satisfies the expected result and no blocking defect prevents completion. It will be marked Fail when the observed result does not satisfy the expected result or a blocking defect prevents the required function.

Criterion	Pass Condition	Requirement
Functional test coverage	All FR-01 to FR-15 test cases are executed before final acceptance.	Mandatory
High-priority functions	All High-priority requirements pass, or an approved exception is documented.	Mandatory
Critical defects	No unresolved Critical defect at acceptance.	Mandatory
Major defects	No unresolved Major defect that prevents a core workflow or compromises security/data integrity.	Mandatory
Security	Unauthorized access to protected student/admin functions is prevented.	Mandatory
Data integrity	No unresolved defect causes loss or corruption of core profile/test data.	Mandatory
Acceptance	Representative end users can complete core student workflows without a blocking issue.	Mandatory
Performance	Critical workflows meet the agreed test-environment response threshold; any breach is documented.	Mandatory
Regression	Previously passed high-priority cases remain passed after approved fixes.	Mandatory

9. TEST DELIVERABLES

- Software Test Plan.
- System feature and quality-attribute requirement list.
- System/Integration test cases and execution records.
- Acceptance test cases and user feedback.
- Unit-test evidence/turnover documentation from developers.
- Defect/incident reports and defect summaries.
- Test logs and test turnover reports.
- Test data and environment setup notes.
- Final test summary with pass/fail status and outstanding defects.

10. STAFFING AND TRAINING NEEDS

The project has two students. Where a separate full-time independent tester is not available, the test lead will coordinate testing with development-team participation. Peer review and role separation will be used where practical.

Role	Assigned Person	Main Duties
Test Lead / QA	Wafa	Prepare test plan, design/execute tests, maintain defect log, coordinate acceptance and prepare final test summary.
Developer / QA Member	Muhammad Talha Islam	Perform unit testing, support integration testing, fix defects and participate in regression testing.
Representative End Users	Selected students	Perform acceptance workflows and provide usability feedback.

Training Needs

- Developers/testers should understand the system requirements and expected student/admin workflows.
- Team members should be familiar with Postman, browser Developer Tools and basic database verification.
- Testers should understand test-case execution, defect severity/priority and regression testing.
- If automation is used, the assigned member should receive basic Selenium/Playwright training.
- The team should maintain the testing schedule using Asana or another project-management tool.

11. RESPONSIBILITIES

This table defines the responsibilities of each role involved in the testing process. It clarifies the duties of the Test Lead, Developer, Project Team, End Users, and Project/Course Reviewer.

Role	Responsibilities
Test Lead / QA	Own test plan; design tests; coordinate execution; review results; maintain defect/incident records; prepare final test summary.
Developer	Perform unit testing; provide testable builds; correct assigned defects; provide unit-test evidence; support system/integration testing.
Project Team	Review requirements; prepare test data; participate in system/integration and regression testing; maintain project documentation.
End Users	Execute representative acceptance tests and provide feedback on usability and suitability.
Project/Course Reviewer	Review the submitted test plan and project evidence as required by the course process.

12. TESTING SCHEDULE

This Gantt chart presents the planned timeline and sequence of major testing activities, from requirement review and test planning to acceptance testing and final report sign-off.

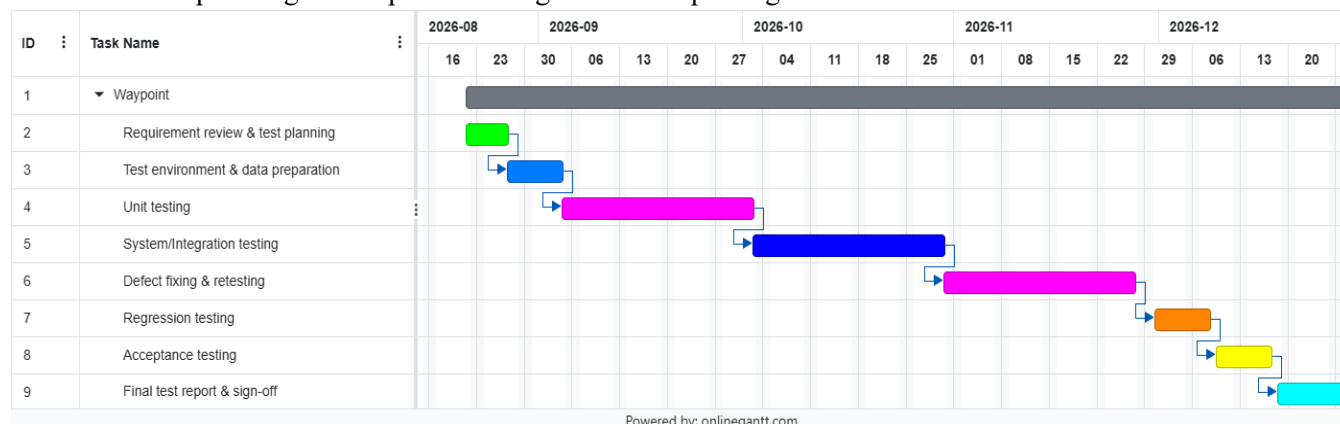


Fig 15: Gantt Chart for Testing Activities

13. PLANNING RISKS AND CONTINGENCIES

This table identifies the major risks that may affect the testing process along with their impact, likelihood, and planned contingency measures.

ID	Risk	Impact	Likelihood	Contingency
R1	Limited testing time	High	Medium	Prioritize high-priority requirements and start system testing early.
R2	Only two team members	High	High	Use peer review and separate development/testing responsibilities where practical.
R3	AI API unavailable	High	Medium	Use controlled mock/stub responses and predefined test data for repeatable tests.
R4	Inconsistent AI recommendations	High	Medium	Use predefined student profiles and a documented expected-output/relevance review method.
R5	Resume parsing problems	Medium	Medium	Test multiple supported/invalid files and retain negative test cases.
R6	Database/test-data corruption	High	Low	Use a dedicated test database, backup/reset data and controlled test records.
R7	Browser/device incompatibility	Medium	Medium	Test representative supported browsers and responsive screen sizes.
R8	Late defect discovery	High	Medium	Perform incremental testing and retest immediately after major fixes.
R9	External job/course/university data unavailable	Medium	Medium	Use controlled mock records while testing the platform interface and data handling.
R10	Independent/industry tester unavailable	Low	High	Use peer review and actual end users; industry checking remains optional under the template.

14. APPROVALS

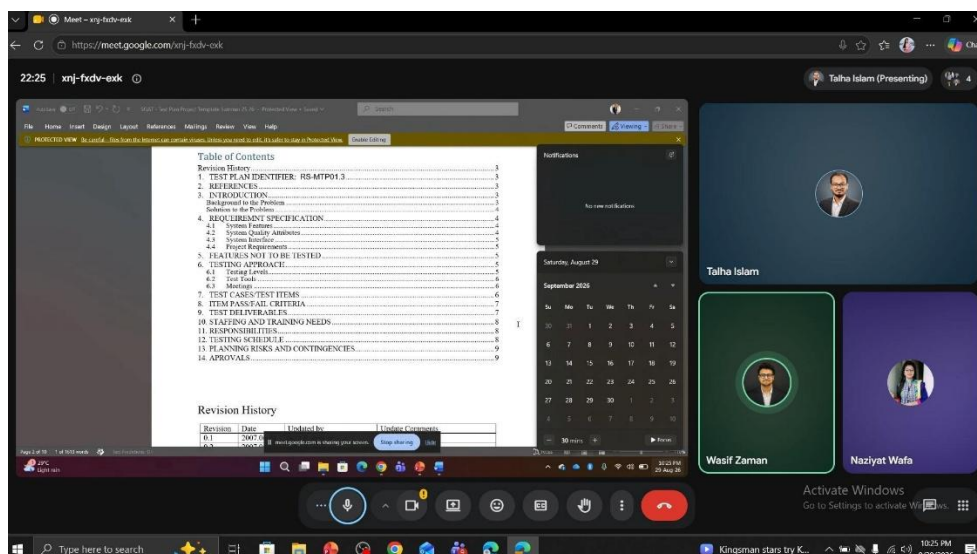
This table lists the designated roles and names responsible for reviewing, managing, and approving different aspects of the project.

Role	Name
Project Sponsor	AIUB
Development Management	Wafa
EDI Project Manager	Muhammad Talha Islam
RS Test Manager	Dale Tester
RS Development Team Manager	Dale Tester
Reassigned Sales	Cathy Stales
Order Entry EDI Team Manager	Julie Order

15. Meetings

1) 29-08-2026 (Online Meeting 1 via Google Meet)

The first introductory meeting was conducted with Wasif Zaman Sunan Bhaiya, SQA Team Lead at Riseup Labs. The meeting focused on discussing the purpose and objectives of the project, clarifying the initial project direction, and establishing effective communication and coordination for future collaboration.



Screenshot 1: Online Meeting

2) 10-09-2026 (Final Meeting)

The final meeting was conducted with Wasif Sunan Bhaiya. During the meeting, he thoroughly reviewed and verified our complete project paper, provided necessary feedback, and officially signed the document upon successful verification.



Screenshot 2: Physical Meeting at Riseup Labs